INTERNSHIP REPORT ON AI-POWERED LEARNING MANAGEMENT SYSTEM (NEELA LMS)

An Internship Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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GITAM
DEEMED TO BE UNIVERSITY

Department of Artificial Intelligence and Data Science
GITAM School of Computer Science and Engineering
GITAM (Deemed to be University)
Visakhapatnam
2026

Department of Artificial Intelligence and Data Science

GITAM School of Computer Science and Engineering
GITAM (Deemed to be University)
Visakhapatnam

GITAM
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the internship report entitled AI-POWERED LEARNING MANAGEMENT SYSTEM (NEELA LMS) is an original work done and submitted to the Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University) in partial fulfillment of the requirements for the award of B.Tech Degree in Computer Science and Engineering (Artificial Intelligence and Machine Learning).

The work has not been submitted to any other college or University for the award of any degree or diploma.

Date: 06.11.2026

Registration No	Name	Signature
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Visakhapatnam

GITAM
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the internship report entitled "AI-POWERED LEARNING MANAGEMENT SYSTEM (NEELA LMS)" is a bonafide record of work carried out by Venkata Sai Satya Mrudula Mahadasyam (2023000597) at 5 Elements Intellli Pvt. Ltd., Visakhapatnam and submitted in partial fulfillment of requirements for the award of B.Tech Degree in Computer Science and Engineering (Artificial Intelligence and Machine Learning).

Date : 6.11.2026

Internship Reviewer

Head of the Department

Head of the Department
Dept. of Artificial Intelligence and Data Science
GITAM School of Computer Science and Engineering
GITAM (Deemed to be University)
Visakhapatnam - 530045

Basic Information of the Internship Report

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S.No.	Description	Details
	Regd. No.	2023000597
	Name of the Student	Venkata sai satya mrudula mahadasyam
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	Type (Govt./Private/Educational/ Training)	Private
	Address of the Organisation	1-12-3 H NO 38, SBI COLONY, PEDA WALTAIR Visakhapatnam - 530017, Andhra Pradesh, India & The Deck , Siripuram, Visakhapatnam
	State in which you worked	Andhra Pradesh
	Website of the Organisation	https://www.5ei.ai/
	Employee Size in the Organisation	10-15
	Contact Details of the Organisation (email id & Contact Number)	contact@5ei.ai , +91 9866768368
	Name of the Mentor from the Organisation	Mr.Shiva Kumar Buddaraju
	Email Id & Contact Number of the Mentor from the Organisation	8247315604, shiv.buddharaju@5ei.ai

	Internship Title .	AI intern
	Skills or Technologies Used	Generative AI, LLMs, RAG, Embeddings, Manual Testing, Playwright, Prompt Engineering, Docker & DevOps
	Internship Mode (Physical / Online / Hybrid)	Hybrid
	Type (Paid by Company/ Free/ Paid by Student)	Free
	Amount (in Rs.)	0
	Internship Duration (Days)	45 days

ACCEPTANCE LETTER/OFFER LETTER

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5ei 5 ELEMENTS INTELLI PRIVATE LIMITED

5 May 2026

OFFER LETTER

Subject: Internship Offer Letter - AI Intern

Dear Ms. Satya Mrudula Mahadasyam,

We are pleased to confirm your selection for the position of AI Intern at 5 Elements Intelli Private Limited.

Your internship is scheduled to commence from 5th May 2026. During this period, you will be working with the Development team, where you will gain hands-on experience and contribute to ongoing projects.

Below are the details of your Internship :

- Role: AI Intern
- Start Date: 5th May 2026
- Duration: 3 months
- Working Hours: Flexible
- Mode: Hybrid

Responsibilities :

- Learn and understand AI and Machine Learning concepts.
- Assist the team in day-to-day project activities.
- Work on assigned AI-related tasks and projects.
- Collect and organize data for analysis.
- Support the development and testing of AI solutions.
- Prepare project documentation and reports.
- Attend team meetings and training sessions.
- Complete tasks assigned by the mentor or reporting manager.

You will be added to the official internship WhatsApp group shortly for regular updates and communication. The necessary training courses and learning materials to support your internship will also be shared with you through the group.

Kindly acknowledge the receipt of this offer and confirm your acceptance at the earliest via mail.

We look forward to having you onboard and wish you a productive and enriching internship experience with us.

A certificate will be issued upon successful completion of the internship.

Buddhiraju Sivakumar

B. Sivakumar
Managing Director

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contact@5ei.ai

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Visakhapatnam - 530017, Andhra Pradesh, India

ABOUT THE COMPANY

5 Elements Intelli Private Limited (5EI) is an AI-first technology company focused on building intelligent enterprise platforms for complex and regulated industries. The company develops platforms that integrate artificial intelligence, unified data, intelligent agents, and governance frameworks to transform complex business systems into scalable, compliant, and outcome-driven solutions.

The company follows an **AI-First approach**, where intelligence is embedded directly into workflows, decision-making, and business operations rather than being added as an afterthought. Its platforms are designed to help organizations augment decision-making, automate processes, optimize operations, and continuously improve their workflows.

A key focus of 5EI is **Unified Data**, which connects and harmonizes information across different enterprise systems to provide a single source of truth. This enables both employees and AI systems to access consistent and contextual information.

The company also emphasizes **Governance by Design**, incorporating compliance, control, auditability, and security into its platforms. This is particularly important for organizations operating in regulated environments where trust, accountability, and long-term sustainability are essential.

5EI works across key sectors including **Agriculture, Healthcare, Finance, and Education**. Its solutions include AI-driven crop and supply-chain insights, intelligent healthcare workflows, financial risk and compliance solutions, and personalized learning and educational intelligence.

One of its education-focused platforms is **NEELA**, which provides AI-powered learning experiences through intelligent tutoring, adaptive assessments, and learning analytics. Through its AI-first platforms, 5 Elements Intelli aims to transform complex enterprise systems into scalable, governed, and intelligent solutions.

INTERNSHIP CERTIFICATE

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5ei 5 ELEMENTS INTELLI PRIVATE LIMITED

20 June 2026

INTERNSHIP COMPLETION CERTIFICATE

Subject: Regarding Completion of Internship

This is to certify that Ms. Satya Mrudula Mahadasyam, a student of B.Tech CSE (AI/ML), has successfully completed an internship as an AI Intern at 5 Elements Intelli Private Limited from 5th May 2026 to 20th June 2026.

During the internship period, the intern was involved in AI-related projects and activities, including research, data analysis, model evaluation, and project documentation. The intern demonstrated dedication, professionalism, and a strong willingness to learn throughout the internship tenure.

We found the performance and conduct of the intern to be satisfactory and appreciate their contribution to the organisation.

We wish her every success in future academic and professional endeavours.

Buddharaju Sivakumar

B. Sivakumar
Managing Director

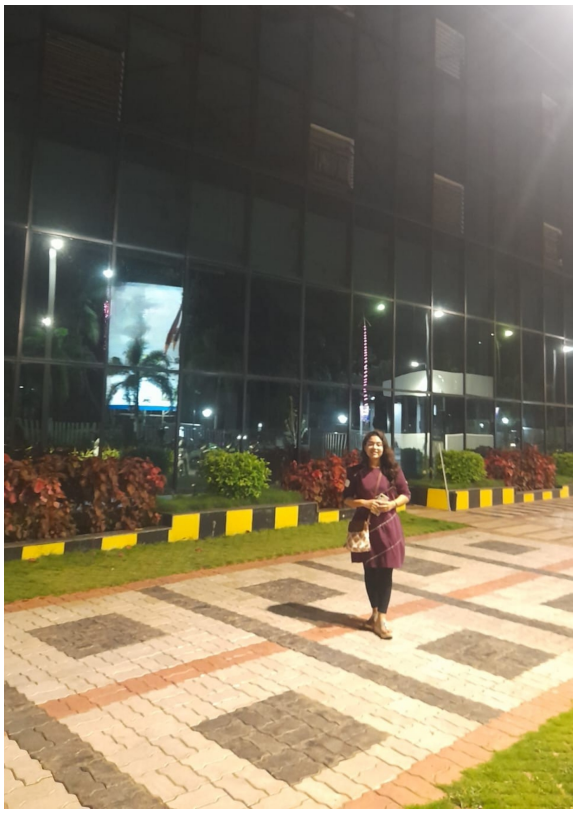
This document is issued electronically on the official letterhead of the organization and does not require a physical signature for validity or authenticity.

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GEO-TAGGED PHOTOGRAPHS

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **GITAM (Deemed to be University), Visakhapatnam**, for providing me with the opportunity to undertake this internship as part of my academic journey. The experience allowed me to move beyond theoretical learning, gain practical exposure, and understand how academic concepts are applied in a professional working environment.

I extend my heartfelt thanks to **Prof. S. Arun Kumar, Dean, GITAM School of Computer Science and Engineering (GSCSE)**, and **Prof. K. Thirupathi Rao, Director, GSCSE**, for their support and for providing an academic environment that encourages learning, innovation, and professional development.

I would also like to express my sincere gratitude to **Dr. K. Naveen Kumar, Head of the Department of Artificial Intelligence and Data Science (AI&DS)**, for his guidance, encouragement, and support throughout this internship. I am equally thankful to the **Department of Artificial Intelligence and Data Science (AI&DS)**, which caters to undergraduate students of CSE – Artificial Intelligence and Machine Learning, Data Science, Internet of Things, and Cyber Security, as well as postgraduate students of Data Science. The department has provided me with a strong academic foundation and encouraged me to explore emerging technologies and their practical applications.

I am grateful to **5 Elements Intelli Pvt. Ltd.**, for providing me with the opportunity to work on the AI-Powered Learning Management System (NEELA LMS). I sincerely thank my mentors and colleagues for their guidance, valuable feedback, and support throughout the internship. Their assistance helped me understand professional workflows, overcome technical challenges, and improve my ability to work on practical tasks.

This internship has been a valuable learning experience, helping me strengthen my technical knowledge, improve my problem-solving skills, and develop a better understanding of teamwork, communication, adaptability, and professional responsibilities. It also gave me the confidence to explore unfamiliar technologies and apply my knowledge to real-world projects.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of my internship and this report. I am truly grateful for their support, encouragement, and the valuable learning opportunities provided throughout this experience.

ABSTRACT

This internship was undertaken at **5 Elements Intelli Private Limited** as an **AI Intern**, providing practical exposure to the application of Artificial Intelligence and Generative AI in a real-world environment. The internship focused primarily on understanding and working with **Neela LMS**, an AI-powered Learning Management System designed to provide intelligent and interactive learning experiences for students and educators.

During the internship, I gained practical understanding of **Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, and prompt engineering**, and explored how these technologies can be integrated into educational applications. Neela LMS incorporates AI-powered academic assistance, intelligent question answering, content retrieval, personalised learning support, and role-based access to enhance the learning experience.

Along with exploring Generative AI concepts, I was involved in **manual testing, test case design, bug reporting, documentation, and basic Playwright automation**. These activities provided practical experience in evaluating application functionality, identifying issues, documenting defects, and understanding the importance of quality assurance in AI-driven systems. I also worked with different user personas and explored GenAI-based solutions to understand how the platform could address different user requirements.

The internship helped bridge the gap between academic concepts and professional implementation by providing hands-on exposure to AI technologies, software testing, automation, documentation, and collaborative work. It strengthened my technical understanding of Generative AI and AI-based applications while also improving my problem-solving, communication, teamwork, and professional skills.

Overall, the internship was a valuable learning experience that enhanced my understanding of how **AI and Generative AI technologies can be applied to real-world platforms**. It provided a strong foundation for further learning and future work in Artificial Intelligence, Generative AI, and intelligent software systems.

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1. OBJECTIVES AND SCOPE

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1.1 Internship Objectives

The main objective of this internship was to get practical experience in Artificial Intelligence and understand how AI concepts are used in real projects. The internship also helped me improve my understanding of Generative AI, learn how AI-based applications are tested, and become familiar with the work involved in a professional development environment.

The key objectives of the internship were:

- To understand the basics of Artificial Intelligence, Machine Learning, and Generative AI.
- To gain hands-on experience with AI-based applications and real-world projects.
- To understand concepts such as LLMs, RAG, embeddings, and prompt engineering.
- To learn how to create and execute test cases for an application.
- To identify, report, and document bugs during testing.
- To get basic exposure to Playwright automation.
- To improve problem-solving, communication, documentation, and teamwork skills.

1.2 Project Objectives

The main project objective was to understand and contribute to the testing and evaluation of **Neela LMS**, an AI-powered learning platform. The work focused on understanding how its AI features respond to different user requirements and checking whether the application works as expected.

The project objectives included testing different features and user personas, understanding AI-generated responses, identifying issues, preparing test cases, and documenting the results. This helped in getting a practical idea of how AI-based applications are evaluated before being used by end users.

1.3 Scope of Work

The scope of my internship mainly covered **AI application testing and Generative AI-related work**. My work included exploring the Neela LMS platform, understanding its AI

features, testing different user scenarios, preparing test cases, reporting bugs, and maintaining relevant documentation.

I also gained basic exposure to **Playwright automation** and learned how automation can be used to support software testing. Along with the technical work, the internship involved working with mentors and team members, discussing issues, understanding requirements, and improving solutions based on feedback.

The internship provided a practical understanding of the overall process involved in working with an AI-based application, from understanding its features and user requirements to testing, identifying problems, and documenting the findings.

2. PROJECT DESCRIPTION

2.1 Problem Statement

Neela LMS is an AI-powered Learning Management System designed to provide students and educators with a more interactive learning experience. Since the platform uses Generative AI for features such as academic assistance and question answering, it is important to make sure that these features work properly for different users and situations.

My internship work focused on understanding the AI features of the platform and helping with their testing and quality evaluation. This included checking different user scenarios, observing AI-generated responses, identifying issues, and documenting the findings.

2.2 Proposed Solution

Neela LMS uses **Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and embedding models** to provide academic assistance and retrieve relevant learning content. The platform supports features such as AI-powered question answering, personalised learning support, course and resource management, and content retrieval.

My contribution was mainly focused on testing these AI-driven features from a user's point of view. I worked with different personas and scenarios to understand whether the expected functionality and AI responses were being delivered correctly.

2.3 Project Overview

As an **AI Intern**, my work on the Neela LMS project involved both **AI-related exploration and quality assurance**. I first familiarised myself with the platform, its features, and the different types of users interacting with it.

My specific contributions included:

- Exploring and understanding the AI features available in Neela LMS.
- Working with different **user personas** and testing their respective workflows.
- Testing AI-powered features and checking the relevance and behaviour of generated responses.
- Preparing and executing **manual test cases** for different features and scenarios.
- Identifying unexpected behaviour and documenting **bugs and issues**.
- Reporting bugs with relevant details so that they could be investigated and addressed.
- Maintaining testing-related documentation and recording observations.
- Gaining basic hands-on exposure to **Playwright automation** for testing.
- Exploring **Generative AI concepts, LLMs, RAG, embeddings, and prompt engineering** in relation to the platform.
- Working with the team and using feedback to improve my understanding of the application and its requirements.

This work helped me understand that testing an AI-based application is different from testing only conventional software features. Along with checking whether a feature works, it is also important to consider the quality, relevance, and consistency of AI-generated responses.

2.4 Expected Outcomes

The main expected outcome of my work was to help improve the reliability and usability of the Neela LMS platform by identifying issues during testing and documenting them clearly. The testing activities also helped evaluate different user scenarios and AI-based features.

From my internship, I gained practical experience in **AI application testing, Generative AI, LLMs, RAG, embeddings, prompt engineering, test case design, bug reporting, documentation, and basic automation**. The experience also helped me improve my

problem-solving, communication, teamwork, and ability to work with unfamiliar technologies.

3. ROLES AND RESPONSIBILITIES

3.1 Assigned Role

I worked as an **AI Intern at 5 Elements Intelli Private Limited**. My primary work was related to the **Neela LMS**, an AI-powered learning management platform. My role involved understanding the AI-based features of the platform and contributing to their testing and evaluation. Along with AI-related learning, I was also involved in manual testing, test case preparation, bug reporting, documentation, and basic test automation.

3.2 Responsibilities

My major responsibilities during the internship included:

- Understanding the **Neela LMS** platform, its features, workflows, and user personas.
- Exploring AI and Generative AI features used within the platform.
- Developing an understanding of **LLMs, RAG, embeddings, and prompt engineering** in the context of the application.
- Testing different features using different **user personas and real-world scenarios**.
- Preparing test cases based on feature requirements and expected behaviour.
- Executing manual test cases and recording the results.
- Testing AI-generated responses and checking whether the responses were relevant and appropriate for the given inputs.
- Identifying functional issues and unexpected behaviour during testing.
- Reporting bugs with proper descriptions and supporting details.
- Maintaining testing and bug-reporting documentation.
- Learning and applying **basic Playwright automation** for testing activities.
- Discussing observations and issues with mentors and team members and incorporating their feedback.

3.3 Tasks Performed

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During my internship, I contributed to the testing and evaluation of the **Neela LMS** platform. I explored its different features and tested workflows from the perspective of different user personas.

A major part of my work involved **manual testing**. I prepared test cases for different scenarios, executed them, and compared the actual results with the expected results. For AI-based features, I tested different inputs and observed the generated responses to check their behaviour and relevance.

I also worked on **identifying and documenting bugs** found during testing. The issues were recorded clearly so that they could be understood and investigated by the development team. This helped me understand the importance of proper bug documentation and communication in software development.

Along with manual testing, I gained practical exposure to **Playwright automation** and understood how automated tests can be used to reduce repetitive testing effort.

I also explored how **Generative AI, LLMs, RAG, embeddings, and prompt engineering** were related to the platform. This helped me connect the AI concepts I had studied academically with their practical use in an actual application.

3.4 Team Collaboration

I regularly interacted with my mentor and team members to understand requirements, clarify issues, discuss testing results, and improve my approach to assigned tasks. I discussed bugs and observations identified during testing and used the feedback provided by the team to improve my work.

Working with different personas and AI-based scenarios also helped me understand how the same application can have different requirements depending on the user. The internship gave me practical experience in **AI application testing, quality assurance, documentation, teamwork, communication, and problem-solving**, while allowing me to contribute to the improvement and evaluation of the Neela LMS platform.

4. TECHNOLOGIES AND TOOLS USED

During my internship, I worked with and learned about different technologies and tools used for understanding, testing, and working with the Neela LMS platform.

4.1 Artificial Intelligence and Generative AI

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- **Large Language Models (LLMs):** Learned the basic concepts of LLMs and their role in AI-powered applications.
- **Generative AI:** Explored how Generative AI can be used to provide academic assistance and generate responses based on user queries.
- **Retrieval-Augmented Generation (RAG):** Learned the basic working of RAG and how retrieved information can be used along with an LLM to generate more relevant responses.
- **Embeddings:** Understood the basic concept of embeddings and their use in content retrieval.
- **Prompt Engineering:** Learned how different prompts and inputs can affect AI-generated responses.

4.2 Frameworks and Libraries

- **LangChain:** Learned the introduction and basic concepts of LangChain to better understand the AI-related project files, their structure, and how different components of an AI application can be connected. This helped me understand the project code and follow the flow of AI-related operations.
- **Playwright:** Used at a basic level to understand and perform automated testing of web application features.

4.3 Testing and Documentation Tools

- **Manual Testing:** Used for testing different features, workflows, user personas, and AI-based functionalities.
- **Test Case Design:** Used to prepare test scenarios based on expected and actual behaviour.
- **Bug Reporting:** Used to record and communicate issues identified during testing.
- **Documentation:** Used to maintain testing observations, test results, and bug-related information.

4.4 Platforms and Project Technologies

The main platform I worked with was **Neela LMS**, an AI-powered Learning Management System that uses technologies such as LLMs, RAG, and embedding models to support intelligent learning and content retrieval.

Through these technologies and tools, I gained practical exposure to both the **AI side of the project** and the **software testing side**, which helped me understand how AI-based applications are developed, evaluated, and improved.

5. METHODOLOGY AND IMPLEMENTATION

5.1 Development Methodology

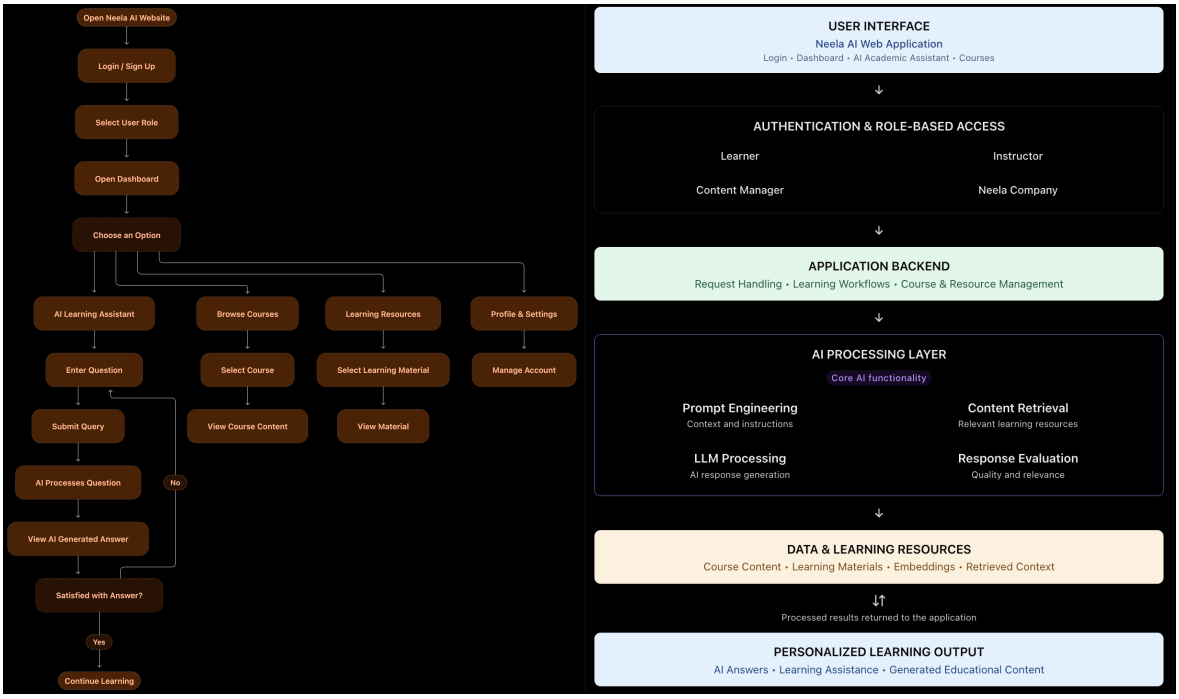
The internship work followed a practical and iterative approach. I first understood the Neela LMS platform, its features, user personas, and requirements. After understanding the application, I explored the AI-related concepts involved and then performed testing based on different user scenarios.

The general workflow followed during my internship was:

Understand the Feature → Understand the AI Component → Prepare Test Scenarios → Execute Tests → Observe Results → Identify Issues → Report and Document → Retest

This approach helped me understand the application step by step and improve my testing process based on the results and feedback received from the team.

5.2 System Architecture



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Neela LMS is an AI-powered learning platform that uses **Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and embedding models** to provide intelligent learning support and content retrieval.

At a high level, the AI workflow can be understood as:

User Query → Application → Relevant Content Retrieval → AI/LLM Processing → Generated Response → User

My role was mainly on the application testing side of this workflow. I interacted with the platform through different user scenarios and evaluated whether the expected functionality and AI-generated responses were being produced correctly.

5.3 Module Design

During the internship, I worked with different functional and AI-related areas of the Neela LMS. These included:

- **User and Persona-based Workflows:** Testing the application from the perspective of different users.
- **AI Academic Assistance:** Testing AI-based interactions and responses to user queries.
- **Content Retrieval:** Understanding and testing features related to retrieving relevant learning content.
- **Learning Features:** Exploring features related to courses, resources, and personalized learning support.
- **Testing and Bug Reporting:** Preparing test cases, executing them, identifying issues, and documenting bugs.

5.4 Implementation

My implementation-related work mainly involved the **testing and evaluation of the existing Neela LMS application** rather than developing the complete platform from scratch.

I first explored the project and its files to understand how the application was organised. As part of this process, I learned the **basics of LangChain** so that I could better understand the AI-related project files and the flow of components involved in the application.

I then worked on manual testing by creating and executing test cases for different features and user scenarios. For AI-based functionalities, I used different inputs and prompts and observed the generated responses. I compared the observed behaviour with the expected behaviour to identify inconsistencies or issues.

When an issue was found, I documented the bug with the necessary details and communicated it to the team. I also gained basic exposure to **Playwright** and understood how automated testing could be used for web application workflows.

5.5 Integration

The different activities performed during the internship were connected as part of the overall testing process. Understanding the AI concepts helped me better understand the features being tested, while manual testing helped identify functional and AI-response-related issues.

The knowledge gained about **LLMs, RAG, embeddings, prompt engineering, and LangChain** helped me understand the AI side of the project. At the same time, **test case design, bug reporting, documentation, and Playwright** provided practical exposure to the quality-assurance side.

This combination helped me understand how AI-based features are integrated into a real-world learning platform and how they can be tested and improved through systematic evaluation.

6. TEST CASES

6.1 Testing Approach

Testing was carried out mainly through **manual testing** of the Neela LMS platform using different user personas and scenarios. The testing focused on checking whether the application features worked according to their expected behaviour and whether AI-based features produced relevant responses.

For AI-related functionalities, different inputs and prompts were used to observe the generated responses. Functional issues and unexpected behaviour were identified and documented for further investigation. Basic **Playwright automation** was also explored as part of the testing process.

6.2 Test Environment

The testing was performed on the **Neela LMS web application** using a web browser. Different user personas and workflows were used to test the application's functionality. AI-related features were tested using different queries and inputs to observe their responses.

The main testing activities included:

- Functional testing of application features
- User persona and workflow testing
- AI response testing
- Test case execution
- Bug identification and reporting
- Retesting after issue resolution
- Basic Playwright-based automation

6.3 Test Cases

The following are representative test cases based on the testing activities performed during the internship:

Test Cas	Test Scenario	Expected Result	Actual Result	Status
TC-01	Login using valid user credentials	User should be successfully logged in and redirected to the	Application behaviour was checked against the	Pass
TC-02	Login using invalid credentials	Application should reject invalid credentials and display	Invalid login behaviour was verified	Pass
TC-03	Submit a valid academic query to	AI assistant should provide a relevant response	Response was generated and evaluated	Pass
BUG-017	Notebook / Source Upload - Page becomes	The page should remain interactive after the upload completes.	The page became unresponsive, preventing further interaction.	Critical
BUG-026	Notebook / Archive Files - Archived file fails to open	Archived files should open and display notebook content.	An “Unable to Connect to API Server” error appeared instead of the archived content.	Critical
TC-06	Access features according to	User should be able to access the features available to that	Persona-based workflows were tested	Pass
TC-07	Access learning resources/content	Relevant learning content should be displayed or	Content-related functionality was	Pass
TC-08	Test an unexpected or invalid input	Application should handle the input without unexpected	Application response was observed and	Pass/ Issue

TC-09	Execute a repeated workflow using	Automated test should perform the defined workflow	Basic Playwright automation was explored	Pass
BUG-031	Ask & Search / Ask Toggle - Answer generation fails because of model configuration	The system should generate an answer based on notebook sources.	An “Ask failed” error indicated that the configured model `test-default-chat-model` could not be found.	Critical

Note: The table represents the types of test scenarios handled during the internship. Individual test results may vary depending on the feature, test data, and application state at the time of testing.

6.4 Test Results

The testing activities helped identify functional issues, unexpected behaviour, and areas requiring further attention in the application. Testing different user personas helped verify the application from different user perspectives, while testing AI-based features provided practical exposure to evaluating generated responses.

The identified issues were documented and communicated to the team for further investigation. The testing process also helped me understand the importance of systematic test case preparation, clear bug reporting, proper documentation, and retesting.

Through this work, I gained practical experience in **manual testing, AI response evaluation, test case design, bug reporting, documentation, and basic Playwright automation**.

7. RESULTS AND DELIVERABLES

7.1 Results

The internship provided me with practical experience in working with an **AI-powered Learning Management System** and understanding how Artificial Intelligence and Generative AI can be applied in a real-world application.

Through my work on **Neela LMS**, I contributed to the testing and evaluation of different application features and user workflows. I tested AI-powered functionalities such as the academic assistant and intelligent question answering, along with features related to content retrieval, personalized learning, course and resource management, authentication, role-based access, and the interactive user interface.

I also tested different user personas and scenarios to understand how the platform behaved for different types of users. During AI feature testing, I used different queries and prompts to observe the generated responses and identify unexpected behaviour.

The internship also helped me gain practical understanding of **LL**Ms, RAG, embeddings, prompt engineering, and LangChain, along with software testing practices such as manual testing, test case design, bug reporting, documentation, and basic Playwright automation.

7.2 Project Deliverables

The major deliverables and outputs from my internship work included:

- Manual test cases for different Neela LMS features and user scenarios.
- Test execution and recording of observations.
- Testing of AI-generated responses using different queries and prompts.
- Identification and documentation of application issues and bugs.
- Bug reports communicated to the concerned team for investigation.
- Testing-related documentation and observations.
- Basic Playwright automation exposure for web application workflows.
- Understanding of the AI-related project structure and files through an introduction to LangChain.
- Practical understanding of how LLMs, RAG, embeddings, and Generative AI are used within an AI-powered learning platform.

7.3 Performance and Output

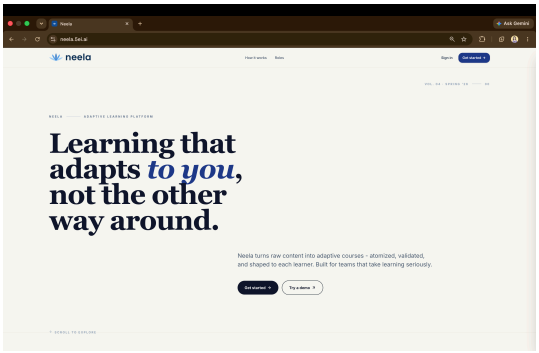
The testing activities helped identify functional and AI-related issues and provided observations that could be used for further improvement of the platform. Testing different personas and scenarios helped provide a broader view of the application's behaviour.

My work also resulted in improved understanding of how AI-based applications need to be evaluated beyond basic functionality. In particular, testing AI features required observing the relevance and behaviour of generated responses for different inputs.

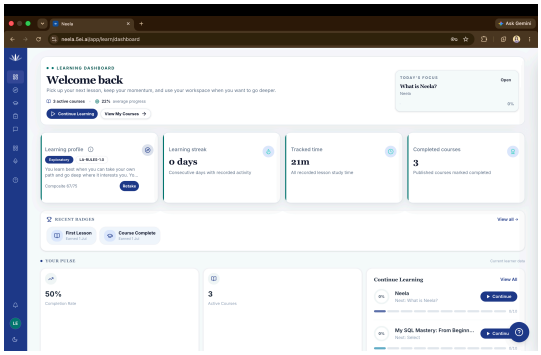
The internship helped me move from theoretical knowledge to practical application and improved my ability to analyse features, create test scenarios, identify issues, document findings, and communicate them effectively.

7.4 Screenshots

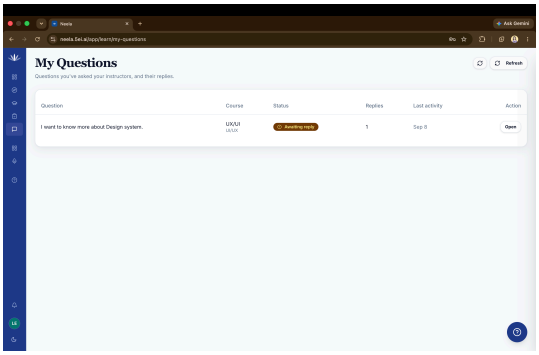
Screenshots of the Neela LMS platform, testing activities, relevant application features, and other internship work can be included in this section as supporting evidence.



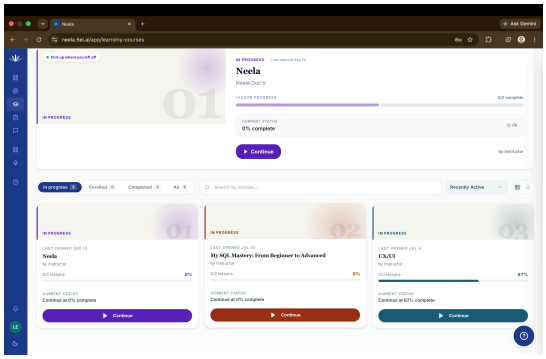
Landing Page



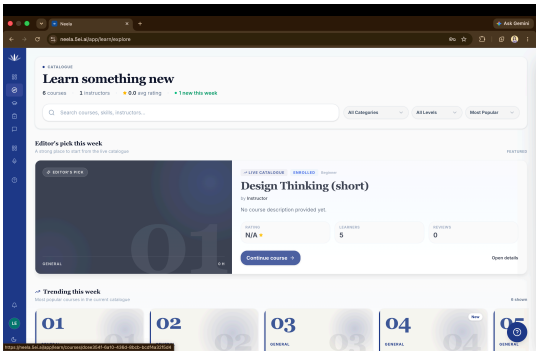
Dashboard



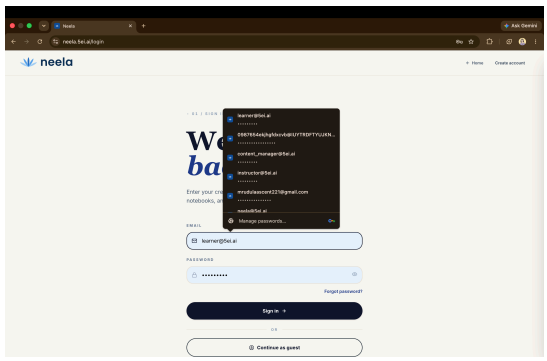
ChatBot questions History



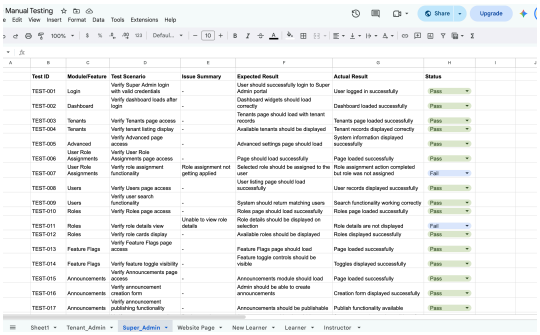
Course content & progress

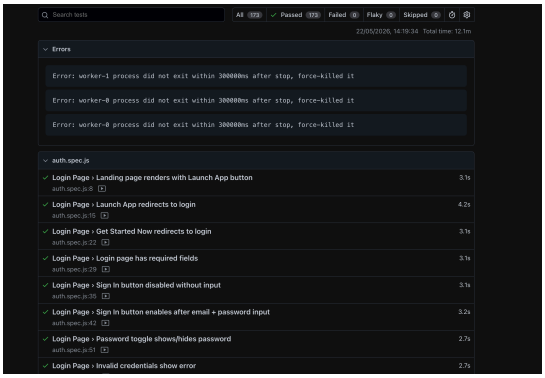


Course Selection Section



Login page & Sign Up





Manual testing **Playwright (Automation)**

8. SKILLS AND LEARNING OUTCOMES

During the internship, I gained practical knowledge of Artificial Intelligence and Generative AI while working with the Neela LMS platform.

Technical Skills:

- LLMs, Generative AI, RAG, embeddings, and prompt engineering.
- Introduction to LangChain for understanding AI-related project files and workflows.
- Manual testing, test case design, bug reporting, and documentation.
- Basic Playwright automation.

Other Learning Outcomes:

- Improved problem-solving and analytical skills.
- Better understanding of testing AI-based applications.
- Improved communication and teamwork through interaction with mentors and team members.
- Gained practical exposure to working in a professional environment.

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The internship at 5 Elements Intelli Private Limited provided me with valuable practical experience in **Artificial Intelligence, Generative AI, and AI application testing**. Working

on Neela LMS helped me connect concepts such as LLMs, RAG, embeddings, prompt engineering; and LangChain with a real-world application. I also gained hands-on experience in manual testing, test case preparation, bug reporting, documentation, and basic Playwright automation.

9.2 Limitations

My work was mainly focused on **testing, evaluation, and understanding the AI-related components** of the existing Neela LMS rather than developing the complete platform.

9.3 Future Scope

The platform can be further improved through better AI response evaluation, enhanced personalization, expanded automation testing, improved content retrieval, and additional intelligent learning features.

10. REFERENCES

1. 5 Elements Intelli Private Limited — Internal training and project resources.
2. Neela LMS — Project documentation and application resources.
3. Playwright — Official documentation.

17 UN Sustainable Development Goals (UN-SDG)

PROJECT PROFORMA **(Keep tick in the appropriate classification)**

Classification of the Project	Application	Product	Research	Review
	✓			

Project Course Outcomes	
Outcome 1	Formulate specific problem statements for well-defined real-life problems with reasonable assumptions and constraints.
Outcome 2	Perform literature search and / or patent search in the area of interest.
Outcome 3	Conduct experiments / Design and Analysis / solution iterations and document the results.
Outcome 4	Perform error analysis / benchmarking / costing, Synthesis the results and arrive at scientific conclusions / products / solution

Outcome 5 .	Document the results in the form of technical report / presentation

CO – PO Mapping

Project Outcomes	Programme Outcomes (POs)												PSOs		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	2	1	3	1				3	3	3		1	2	2
CO 2						2			2	3	2	2	1	2	2
CO 3	1	1	3			3	3						2	2	2
CO 4	2	2	2	3	2	2			1	1	1	3	3	2	2
CO 5	1	1	1	3	2	2							3	2	2

Note: Identify the SDG Goal that maps your internship from the list and fill the below table. A sample is shared.

UN Sustainable Development Goals (UN-SDG) Mapping

No.	Description
SDG 9	Industry, Innovation and Infrastructure
SDG 10	Reduced Inequalities
SDG 4	Quality Education

Project Title	Project Type	SDG Target	Justification
AI-Powered Learning Management System (NEELA LMS)	Internship	Goal 4: Quality Education	The project contributes to SDG 4 by developing an AI-powered learning management system that improves access to digital learning resources and supports learners and instructors through intelligent learning assistance and content management. It aims to make learning more accessible, organized and effective.

GITAM
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Hyderabad

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mandal, Hyderabad -
502329, Telengana, India

Visakhapatnam

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Visakhapatnam- 530045,
Andhra Pradesh, India


Bengaluru


Visakhapatnam

GIMSR
Medical College Rd,
Nagar, Rushikonda,

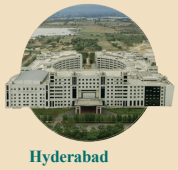
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